

Generating Intermediate Files in GCC

In GCC (GNU Compiler Collection), you can generate all intermediate files during the compilation process using specific options. These intermediate files include preprocessed source files, assembly code, object files, etc.

1. Generate the Preprocessed File (.i or .ii)

Use the `-E` option to stop after the preprocessing step.

```
gcc -E source.c -o source.i
```

2. Generate the Assembly File (.s)

Use the `-S` option to stop after generating the assembly code.

```
gcc -S source.c -o source.s
```

3. Generate the Object File (.o)

Use the `-c` option to stop after the compilation step (object file creation).

```
gcc -c source.c -o source.o
```

4. Generate All Files Sequentially

You can run these commands sequentially to generate all intermediate files for a single source file.

5. Use `-save-temps` to Automatically Save Intermediate Files

If you want GCC to automatically save all intermediate files (preprocessed file, assembly file, and object file), use the `-save-temps` option.

```
gcc -save-temps source.c
```

This will generate the following files in the same directory:

- `source.i` (preprocessed file)
- `source.s` (assembly file)
- `source.o` (object file)

6. Compile and Link (Generate Executable)

If you want to complete the process and generate the executable:

```
gcc source.c -o executable
```

Intermediate files will not be saved unless you use `-save-temps`.